Project Proposal: Attention Mechanisms in Sequence Models for Time-Series Forecasting

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1 Application Area

This project addresses **time-series forecasting**, a fundamental problem spanning energy systems, climate science, and financial markets. Accurate forecasting of temporal signals is critical in each of these domains—grid operators rely on load predictions to schedule generation and maintain stability, meteorologists use atmospheric forecasts to issue warnings and guide planning, and financial institutions depend on exchange rate projections for risk management and portfolio allocation.

Rather than optimizing a forecasting model for a single domain, this project investigates a deeper methodological question: **how do different attention mechanisms in neural sequence models affect forecasting accuracy, computational efficiency, and model interpretability across time-series data with fundamentally different temporal characteristics?** Attention mechanisms—including Bahdanau-style additive attention, self-attention, and cross-attention—have become central to modern sequence modeling since the introduction of the Transformer architecture, yet their relative strengths and tradeoffs for time-series forecasting remain insufficiently understood. Most published work introduces a new architecture and benchmarks it; this project instead holds the broader architecture constant and isolates the effect of the attention mechanism itself, producing a controlled comparative study across three diverse temporal domains that each present distinct challenges: multi-scale infrastructure dependencies, strong environmental periodicity, and stochastic financial volatility.

2 Dataset Selection

This project uses three public time-series datasets deliberately chosen to represent fundamentally different temporal characteristics, enabling a controlled comparison of how attention mechanisms respond to varying data properties.

Dataset 1: Electricity Transformer Temperature (ETTh1)

Source: GitHub—`https://github.com/zhouhaoyi/ETDataset` (introduced in Zhou et al., "Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting," AAAI 2021).

ETTh1 contains approximately 17,420 hourly measurements from an electricity transformer in a province of China, spanning July 2016 to July 2018. It includes 7 features: oil temperature (the primary prediction target) and 6 power load variables representing high/medium useful load and high/medium useless load. The data exhibits multi-scale temporal dependencies—daily load cycles, weekly patterns, and seasonal trends—with complex cross-feature correlations typical of infrastructure systems. No missing values. ETTh1 is the standard benchmark in the transformer-based forecasting literature (Informer, Autoformer, FEDformer), making results directly comparable to published work.

Limitations: Data originates from a single transformer station, limiting geographic generalization. The moderate size (∼17K samples) may constrain deep models if not carefully regularized.

Dataset 2: Jena Climate

Source: Max Planck Institute for Biogeochemistry / Kaggle—`https://www.kaggle.com/datasets/mnassrib/jena-climate`.

The Jena Climate dataset contains approximately 420,000 observations of 14 meteorological variables—air temperature, atmospheric pressure, relative humidity, vapor pressure, wind speed, wind direction, and others—recorded every 10 minutes at a weather station in Jena, Germany from 2009 to 2016. The primary prediction target is air temperature (°C). The data exhibits strong, well-defined periodicity (clear diurnal and annual cycles) with smooth trends and high inter-feature correlation (e.g., temperature, dew point, and humidity are physically coupled). Data will be resampled to hourly resolution to align with ETTh1 and manage computational cost, yielding approximately 70,000 effective samples.

Limitations: Single-station data; resampling discards sub-hourly variation that may contain useful signal for certain forecasting tasks.

Dataset 3: Exchange Rate

Source: GitHub—`https://github.com/laiguokun/multivariate-time-series-data` (used in Lai et al., "Modeling Long- and Short-Term Temporal Patterns with Deep Neural Networks," SIGIR 2018, and adopted as a standard benchmark in the Informer, Autoformer, and FEDformer papers).

The Exchange Rate dataset contains approximately 7,588 daily observations of exchange rates for 8 countries (Australia, United Kingdom, Canada, Switzerland, China, Japan, New Zealand, and Singapore) relative to the US dollar, spanning 1990 to 2016. The data is characterized by high stochastic volatility, limited seasonality, and near-random-walk behavior—properties that make it fundamentally harder to forecast than periodic data. No missing values.

Limitations: Daily frequency means substantially fewer samples than the other two datasets. The near-random-walk nature may limit the absolute accuracy achievable by any model, though relative differences between attention mechanisms remain informative.

The three datasets are deliberately chosen to span the spectrum of temporal characteristics relevant to the research question. ETTh1 presents **complex multi-scale load dependencies with moderate dimensionality** (7 features). Jena Climate offers **strong periodic structure with higher dimensionality** (14 features) and substantially more data. Exchange Rate introduces **stochastic, aperiodic behavior with low dimensionality** (8 features) and the fewest samples. If attention mechanisms respond differently to periodicity, multi-scale structure, and noise-dominated dynamics, this three-dataset design will reveal it.

3 Data Type Characterization

All three datasets share the following properties:

- **Structured**—tabular format with well-defined columns and regular sampling intervals
- **Labeled**—continuous numerical target values available for supervised learning (oil temperature for ETTh1, air temperature for Jena Climate, a selected country's exchange rate for Exchange Rate)

- **Quantitative**—all features are continuous numerical measurements; no categorical or text variables
- **Time-dependent**—multivariate time series sampled at regular intervals (hourly for ETTh1; 10-minute resampled to hourly for Jena Climate; daily for Exchange Rate)

Additionally, all three datasets will undergo statistical characterization prior to modeling—including trend analysis, STL seasonal-trend decomposition, and autocorrelation/partial autocorrelation structure analysis—to formally document the temporal properties that each attention mechanism must learn to exploit. This characterization step directly supports the interpretability analysis by establishing ground-truth temporal structure against which attention weight patterns can be compared.

4 Data Science / Machine Learning Problem Definition

Task type: Multivariate time-series forecasting (supervised regression), with a parallel interpretability analysis of learned attention weight distributions across mechanisms.

Inputs (X): For each prediction, the input is a sliding window of the past L timesteps across all feature channels—a matrix of shape ($L \times d$), where L is the lookback window length and d is the feature dimensionality (7 for ETTh1, 14 for Jena Climate, 8 for Exchange Rate). Temporal position encodings (hour-of-day, day-of-week, month-of-year for the hourly datasets; day-of-week, month-of-year for the daily Exchange Rate data) will be appended as additional input features to provide explicit calendar context to all three model architectures.

Outputs (y): Predicted future values of the target variable at multiple forecast horizons. For ETTh1 and Jena Climate, horizons of 96, 192, 336, and 720 steps ahead will be used, following the standard protocol from the Informer literature. For Exchange Rate, horizons will be scaled to the daily frequency (24, 48, 96, and 192 days ahead). Testing multiple horizons per dataset enables analysis of whether attention mechanisms degrade differently as the prediction window increases.

What success looks like: A strong outcome would achieve four objectives aligned with the research question:

1. **Performance differentiation:** Demonstrate statistically measurable differences in forecasting accuracy (MSE, MAE) between the three attention variants—(a) LSTM with Bahdanau-style additive attention, (b) LSTM with self-attention, and (c) Transformer encoder-decoder with multi-head self-attention and cross-attention—across all three datasets and all forecast horizons.
2. **Interpretability insights:** Generate attention weight visualizations showing which historical timesteps each mechanism attends to, and analyze whether these patterns align with known domain structure (e.g., attention peaks at 24-hour lags for daily cycles in ETTh1/Jena, versus diffuse or uniform attention on the stochastic Exchange Rate data).
3. **Efficiency tradeoffs:** Quantify the computational cost of each mechanism in terms of training convergence speed, inference latency, and total parameter count, producing an accuracy-vs-efficiency comparison across dataset scales.
4. **Empirical guidelines:** Synthesize findings into actionable recommendations for when each attention variant is most appropriate given a dataset's temporal characteristics—specifically, whether strong periodicity, multi-scale dependencies, stochastic volatility, or high dimensionality favor one mechanism over another.

5 Why Data Science / Machine Learning?

Time-series forecasting in the domains covered by this project—energy infrastructure, climate, and finance—involves nonlinear dynamics, multi-scale temporal dependencies, and complex cross-variable interactions that resist simple analytical solutions.

A physics-based approach to transformer oil temperature prediction would require detailed thermal models of the transformer, knowledge of load scheduling, and ambient temperature coupling—and would still fail to generalize across different transformers or operating regimes without extensive recalibration. Atmospheric temperature forecasting from first principles requires solving partial differential equations governing fluid dynamics and thermodynamics at high spatial and temporal resolution—computationally prohibitive for a single weather station's local prediction task. Financial exchange rates are even more resistant to mechanistic modeling: they are driven by macroeconomic policy, geopolitical events, and market sentiment—factors that defy closed-form formulation.

Classical statistical methods (ARIMA, exponential smoothing) can capture linear temporal structure and simple seasonality, but they assume stationarity, struggle with multivariate dependencies, and cannot model the nonlinear interactions that attention mechanisms are specifically designed to learn. A machine learning approach offers the ability to learn complex multivariate patterns directly from data, generalize across operating conditions and forecast horizons, and—critically for this project—provide interpretability through attention weight analysis. Unlike black-box models where prediction logic is opaque, attention mechanisms produce explicit weight distributions over historical timesteps, offering a window into *what the model has learned to focus on.* This interpretability dimension is central to the research question and is not achievable through traditional statistical or rule-based forecasting approaches.

Furthermore, the comparative nature of this project—isolating the effect of the attention mechanism while controlling for architecture, training procedure, and data—is inherently a data-driven empirical exercise. The question of which attention variant works best under which temporal conditions cannot be answered theoretically; it requires systematic experimentation across diverse datasets.

6 Planned Approaches

6.1 Preprocessing / Feature Work

All preprocessing will be applied identically across the three attention model variants to ensure a controlled comparison:

- **Resampling:** Jena Climate data will be resampled from 10-minute to hourly resolution via mean aggregation to align with ETTh1. Exchange Rate data remains at daily frequency.
- **Train/validation/test splits:** ETTh1 and Jena Climate will follow the standard 12-month/4-month/4-month chronological split from the Informer literature. Exchange Rate will use a 70%/10%/20% chronological split consistent with published benchmarks. All splits are strictly chronological to prevent temporal data leakage.
- **Normalization:** Standard scaling (zero mean, unit variance) fit on training data only, applied to validation and test via `transform()`. Scaling parameters stored per dataset for reproducibility.
- **Temporal feature engineering:** Hour-of-day, day-of-week, and month-of-year encodings appended as sinusoidal positional features for the hourly datasets. Day-of-week and month-of-year for the daily Exchange Rate data.

- **Sliding window construction:** Input sequences of length L (tuned per dataset; starting values of 96 for hourly data, 30 for daily data) with target sequences at horizons of {96, 192, 336, 720} for hourly datasets and {24, 48, 96, 192} for Exchange Rate.
- **EDA and statistical characterization:** Prior to modeling, each dataset will undergo trend analysis, STL seasonal-trend decomposition, and autocorrelation/partial autocorrelation (ACF/PACF) analysis. These will be visualized as time-series plots, decomposition figures, and correlograms, establishing the ground-truth temporal structure against which attention weight patterns will later be compared.

6.2 Learning Methods / Models

Three attention mechanism variants will be implemented in PyTorch, each embedded within a sequence-to-sequence forecasting architecture:

1. **LSTM with Additive (Bahdanau) Attention:** A traditional encoder-decoder architecture where an LSTM encoder processes the input sequence and a Bahdanau-style attention mechanism computes a learned weighted combination of encoder hidden states at each decoder step. This serves as the baseline attention model, representing the pre-Transformer paradigm. The attention score function is a single-layer feedforward network over concatenated encoder and decoder states.

2. **LSTM with Self-Attention:** An LSTM encoder augmented with a self-attention layer applied over the encoder's hidden state sequence before passing to the decoder. This tests whether allowing the encoder to attend to its own outputs—capturing long-range dependencies within the input window—improves over the standard additive attention applied only at decode time. The self-attention is implemented as scaled dot-product attention.

3. **Transformer Encoder-Decoder:** A full Transformer architecture with multi-head self-attention in the encoder, masked multi-head self-attention in the decoder, and multi-head cross-attention connecting them. This represents the modern paradigm and tests whether the elimination of recurrence in favor of pure attention improves forecasting performance and interpretability.

Experimental controls: All three models will share identical training procedures—Adam optimizer, same learning rate schedule, same batch size, same early stopping criteria (patience on validation loss), and fixed random seeds. Hidden dimensions and total parameter counts will be matched as closely as possible across architectures to isolate the effect of the attention mechanism rather than model capacity. Dropout will be applied within all architectures to mitigate overfitting, particularly on the smaller ETTh1 and Exchange Rate datasets. Both train and test metrics will be reported side by side to make any overfitting visible.

Hyperparameter tuning: A focused grid search over key hyperparameters (learning rate, hidden dimension, number of attention heads for the Transformer, lookback window length) will be conducted using validation set performance. Tuning will be performed independently per dataset to allow each mechanism to achieve its best-case performance on each temporal regime.

7 Evaluation Plan

Metrics:

- **Primary forecasting metrics:** Mean Squared Error (MSE) and Mean Absolute Error (MAE), reported per model, per dataset, per forecast horizon. These are the standard metrics in the Informer/Autoformer literature, enabling direct comparison with published results.
- **Efficiency metrics:** Training convergence speed (epochs to best validation loss), wall-clock inference latency per prediction, and total trainable parameter count per model.
- **Interpretability metrics:** Attention weight entropy (measuring how concentrated vs. diffuse attention is across timesteps) and peak attention lag analysis (identifying which historical offsets receive highest attention).

Evaluation strategy:

- All metrics reported on held-out test sets using the chronological splits described in Section 6.1. No test data is used during training or hyperparameter selection.
- Results reported with variability: for each configuration, training will be repeated across 3–5 random seeds with metrics reported as mean $\pm$ standard deviation to account for initialization sensitivity.
- Statistical significance of performance differences between attention mechanisms will be assessed to ensure observed differences are not attributable to random variation.
- Primary comparison metric is MSE at each forecast horizon; MAE serves as supporting metric.

Planned visualizations:

- **EDA figures:** Raw time-series plots, STL decomposition (trend/seasonal/residual), and ACF/PACF correlograms for each dataset—establishing ground-truth temporal structure.
- **Horizon-wise performance bar charts:** MSE on the y-axis, forecast horizon on the x-axis, grouped by attention mechanism, one chart per dataset.
- **Training loss curves:** All three models overlaid on the same plot per dataset, showing convergence speed differences and any overfitting divergence between train and validation loss.
- **Summary heatmap table:** Datasets as rows, models $\times$ horizons as columns, cells colored by MSE—providing the full comparison grid at a glance.
- **Attention weight heatmaps:** For representative test samples, 2D heatmaps where the x-axis is the historical timestep being attended to and the y-axis is the prediction step, generated side by side for all three mechanisms.
- **Attention weight vs. ACF overlay:** Average attention weight distribution overlaid on the autocorrelation function for each dataset. If attention peaks align with ACF peaks, the model has learned genuine temporal structure; if they diverge, the attention may be uninformative. This serves as the central interpretability figure.
- **Accuracy-vs-efficiency scatter plot:** MSE on the x-axis, inference latency or parameter count on the y-axis, one point per model per dataset—showing the Pareto frontier of performance vs. computational cost.

8 Team Plan

This is a solo project conducted as an ad hoc departmental honors extension of EE 344. All work—including data preparation, model implementation, experimentation, evaluation, and the final research paper—will be completed individually.